RAJALAKSHMI ENGINEERING COLLEGE

Sudarshan S S 251001271-2025-IT

S2

Started on	Wednesday, 5 August 2026, 6:35 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 6:44 PM
Time taken	9 mins 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

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Falling back to raw text area.

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int count = 0;
    int i = 1;
    count++;
    int s = 1;
    count++;
    while (s<=n) {
        count++;
        i++;
        count++;
        s += i;
        count++;
    }
    count++;
    printf("%d", count);
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Sudarshan S S 251001271-2025-IT

S2

Started on	Wednesday, 5 August 2026, 6:44 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 6:49 PM
Time taken	4 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void func(int n)
{
    int c=0;
    if(n==1)
    {
      c++;
    }
    else
    {
     c++;
     for(int i=1; i<=n; i++)
     {
         c++;
         c++;
       for(int j=1; j<=n; j++)
       {
         c++;
         c++;
          break;
       }
       c++;
     }
     c++;
    }
    printf("%d",c);
 }
 int main()
```

```
29 {
30     int n;
31     scanf("%d",&n);
32     func(n);
33 }
34
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Wednesday, 5 August 2026, 6:49 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 6:55 PM
Time taken	6 mins 14 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
void factor(int n){
    int c = 0;
    for(int i=1;i<=n;++i){
        c++;
        c++;
        if(n%i==0){
            c++;
        }

    }
    c++;
    printf("%d",c);
}

int main(){
    int n;
    scanf("%d",&n);
    factor(n);


}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Sudarshan S S 251001271-2025-IT

S2

Started on	Wednesday, 5 August 2026, 6:55 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 6:56 PM
Time taken	41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
void func(int n){
    int count=0;
    int c=0;
    count++;
    for(int i=n/2;i<n;i++){
        count++;
        count++;
        for(int j=1;j<n;j=2*j){
            count++;
            count++;
            for(int k=1;k<n;k=k*2){
                count++;
                count++;
                c++;
            }
        }
    }
    count++;

    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    func(n);
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Sudarshan S S 251001271-2025-IT

S2

Started on	Wednesday, 5 August 2026, 6:57 PM
State	Finished
Completed on	Wednesday, 5 August 2026, 7:14 PM
Time taken	17 mins 55 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
void reverse(int n){
    int count = 0;
    int rev = 0, remainder;
    while(count++, n != 0){
        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    printf("%d", count + 2);
}

int main(){
    int n;
    scanf("%d",&n);
    reverse(n);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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